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**Maximum Speed Exposures in Australian Rules Football: Do Methods Matter?**

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**Maximum Speed Exposures in Australian Rules Football: Do Methods Matter?**

**Abstract**

This study compared weekly near-to maximum speed exposure data when determined via different methods and criteria. 47 professional Australian Rules Football (AF) players undertook pre-season sprint assessments (PSSA) via 3 x 50m maximal efforts using 10 Hz GPS over two consecutive seasons. The same technology was used continually during the in-season to identify maximum speeds attained in training and matches. Weekly near-to maximal speed (MS) exposure counts were aggregated for speeds ≥80%, ≥85%, ≥90% and ≥95% of their individual maximum for both determination approaches. Weekly near-to MS exposures was lower ($p<0.0001$) when determined from in-season monitoring for ≥80% (-1.26; CI: -1.58 to -0.93), ≥85% (-0.78; CI: -0.97 to -0.59), ≥90% (-0.42; CI: -0.53 to -0.32), and ≥95% (-0.09; CI: -0.12 to -0.06) versus PSSA, with no effect of playing position ($P \geq 0.161$). Although ≥80% and ≥85% near-to maximum speed exposure data was meaningfully influenced by the determination method, the effect was somewhat trivial at higher speed criteria (≥90% and ≥95%) often considered important for performance gains and injury risk reduction purposes. Maximum speed determination methods therefore may be used interchangeably, and discrete sprint assessments may not be necessary for this purpose.

**Key-words:** sprinting, GPS, maximum speed, team-sports, methods

## Introduction

Sprinting in team-sports is often associated with pivotal events deemed crucial to performance outcomes [1]. Indeed, monitoring and prescription of sprint running volumes [2] and near-to maximum speed (MS) exposures is considered a priority in managing athletic loads for the development of physical qualities and to mitigate injury risk. Studies have suggested that appropriate exposure to high speeds (≥80 - ≥95% of maximum) as a protective factor [3] in reducing non-contact hamstring strain injuries which incur a substantial economic burden [4] and performance detriments [5]. The routine implementation of near-to maximal sprint efforts during training are considered to exert a sufficient mechanical stimulus on hamstring muscles to yield a protective effect against strain injury[6].

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Exposures to near maximum speeds during a training week may be important to both improving performance potential and reducing injury risk, but there remains ambiguity around the procedures used to calculate such exposures [3,7,8]. Traditionally, MS has been determined using radar guns or infrared timing gates during maximum sprint efforts as part of individual physical screening assessments [9]. However, improvements in GPS technology and its proliferation have provided a valid [9,10] and pragmatic alternative in professional team sports. Indeed, research in Soccer [11] and Australian Rules Football (AF) [12] have shown higher maximum speeds attained (2-3%) via GPS (10Hz) tracking during matches versus discrete sprint assessments. The effect magnitude of this MS difference has been deemed meaningful by team-sports industry practitioners [13], but how the determination procedure effects the interpretation of near-to MS exposures is unknown.

The varying methods to determine MS and subsequent criteria to categorise exposure counts may have implications for athlete monitoring and injury-risk management. Therefore, the aim of this study was to compare weekly MS exposure data when determined from 10 Hz GPS tracking data in either discrete sprint assessments during the pre-season, versus the highest MS observed via routine tracking of training and matches across the in-season. Since MS observations in team-sports matches are also nuanced by positional role [11,12,14], we also examined its potential impact upon MS exposure data in an AF context. Finally, we examined the impact of different fixed-fractions of MS commonly used on weekly exposure counts.

**Materials and Methods**

Participants

Player data (n=47) from a professional men's AF club was collected over two consecutive seasons (Ethical approval: H12703). Players (Age: 23.7 ± 3.7 yrs; Body Mass: 87.6 ± 8.0 kg; Stature: 189.1 ± 5.9 cm) were categorised into positional groups based on where they played the majority of time during each match: Nomadic (midfielders, running backs and small forwards) and Non-Nomadic (key position and rucks).

Near-to MS Efforts

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Players MS was defined as the highest running speed measured during either 1) Pre-season sprint assessments (PSSA) or 2) during matches and training sessions throughout the in-season. For PSSA, players completed three 50m maximal sprints on a grass track following a thorough warmup. Maximum speed was determined as the highest instantaneous speed via 10Hz GPS (Optimeye S5, Catapult Sports, Melbourne, Australia) and assigned retrospectively for the corresponding season. PSSA was administered during a tapering phase at the conclusion of the pre-season to allow for training adaptations and limit the impact of residual fatigue. In-season MS was also determined via routine 10 Hz GPS tracking during matches (n=585 individual match observations) and all training sessions (n=6099; average 74.6 per player per season), including prescribed linear MS sprinting drills. For both approaches, near-to MS exposures were recorded via the same technology, and aggregated into weekly (match to match) micro-cycles. Each GPS device was assigned to an individual player and after each session, devices were downloaded using proprietary software (Openfield 1.17.1, Catapult Sports, Melbourne, Australia), with GPS velocity derived from doppler-shift data. MS efforts were recorded when signal quality met the inclusion criteria (number of connected satellites ≥9 and horizontal dilution of position [HDOP] <1.25), and the minimum effort duration was set at 0.3 seconds. The mean HDOP was 0.8 ± 0.1 au and the mean satellite availability was 12.0 ± 0.9 at specific instance that a MS effort was identified, with no MS observations excluded prior to analysis.

The main outcome measures were the number of exposures where players attained speeds ≥80%, ≥85%, ≥90% and ≥95% of their maximum recorded either in the PSSA or during in-season monitoring based on previous research in team-sports [3,7,8].

Statistical Analysis

To the model differences in near-to MS exposure counts between determination methods, a zero inflated negative binomial (quadratic assumption; nbinom2) model term was fit using the glmmTmb package [15] in R studio. Specifically, the model included an interaction between the exposure criteria (e.g. 80%, 85% count etc.) and the method used to quantify the exposure (pre-season sprint assessments vs. in-season monitoring). Additionally, an interaction effect between position and speed threshold was also included in the model to account for potential differences that may exist between nomadic and non-nomadic positions, whilst a random

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intercept was also specified to account for repeated observations within individuals. As it was reasonable to expect that more zero effort counts may exist for ≥95% versus ≥80% exposure criteria, a zero-inflation term within the glmmTMB model was assessed via the ziformula argument. Visual inspection of residual plots and zero inflation were performed using the DHARMa package [16] along with information criterion (AIC) statistics across a range of model specifications assessing for zero inflation and dispersion. Contrast between estimates were performed using emmeans package [17] with 95% confidence intervals (CI) and tukey adjusted p-values reported.

**Results**

Maximal Speed occurrences were recorded during in-season training (55%; 32.4 ± 1.4 $km \cdot h^{-1}$) and matches (45%; 33.3 ± 1.3 $km \cdot h^{-1}$), but not in PSSA (0%; 31.0 ± 1.0 $km \cdot h^{-1}$). The number of weekly MS exposures was lower ($p<0.0001$) when determined from in-season monitoring for ≥80% (-1.26; CI: -1.58 to -0.93), ≥85% (-0.78; CI: -0.97 to -0.59), ≥90% (-0.42; CI: -0.53 to -0.32), and ≥95% (-0.09; CI: -0.12 to -0.06) versus PSSA (Figure 1). There was no effect of playing position in the weekly near-to MS exposure counts ($P \geq 0.161$) for ≥80% (1.34; CI: -0.51 to 3.18), ≥85% (0.35; CI: -0.49 to 1.19), ≥90% (-0.06; CI: -0.26 to 0.38), and ≥95% (-0.01; CI: -0.07 to 0.05) (Figure 2).

***Figure 1 around here***

*** Figure 2 around here ***

**Discussion**

The aims of this methodological study were to examine the effects of the criteria applied and the determination methods upon the quantification of near to maximum speed exposures in a team-sports context. We observed lower maximum speed exposures during the weekly micro-cycle when the maximal speed was ascertained from routine tracking of 10Hz GPS data during in-season training and matches, versus discrete sprint assessments conducted during the pre-season. The magnitude of this methodological difference reduced as the criteria ascended proximally to maximum speed. Positional role had little impact upon the

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frequency of maximum speed exposures during the micro-cycle, and was not an impactful factor in the methodological approach to setting maximum speed criteria.

On average, approximately one additional exposure per in-season micro-cycle was observed when using PSSA to anchor maximum speed criteria for ≥80 and ≥85%. Interpreting the practical relevance of this effect magnitude is challenging considering the wide variety of exposures reported in previous team-sport studies [3,8] together with the inherent challenges in identifying injury risk factors in team-sports research (i.e. low number of injuries, multi-factorial nature of injury inciting events). Alternatively, adopting PSSA maximum speed data increased the exposures ≥95% by just ~0.1 per week. This effect magnitude may be deemed trivial considering the previous team-sports research suggesting that 1 exposure per week [7], or 5-6 exposures ≥95% over a 4-week period [8] reduced non-contact lower limb injury risk. Since, exposures in excess of 95% seem to have more relevance from both theoretical framework [6] and injury risk reduction perspective [18], the current findings may suggest that the methods of maximum speed determination examined herein may be used interchangeably, despite the 'practically relevant' [13] differences in maximal velocities observed both here (~1.3 $km \cdot h^{-1}$) and in previous studies (~0.3 $km \cdot h^{-1}$; [11,12]).

This current study observed no impact of playing position upon near-to MS exposures or their means of anchoring. Non-nomadic roles in elite AF are often characterised by larger players and their movement demands are largely dependent upon match situational factors [19]. We hypothesised that these non-nomadic players would have less near-to MS exposures than nomadic players who cover more ground at higher speeds during matches [20], and we expected positional role to nuance the max speed method determination. The null findings may be explained by the heterogeneity within the movement profiles of non-nomadic playing positions [19], and that 55% of MS's were attained in training sessions. Hence, future work may be warranted to determine if particular roles (i.e. ruckmen) influence MS determination approaches and their subsequent exposures.

Considering the applied nature of this study, a number of considerations should influence the interpretation and generalizability of its findings. For pragmatic reasons, we were only able to contrast maximum speed determination measures via 10Hz GPS, whilst this approach has acceptable validity versus criterion measures [9,11,13], higher resolution laser or radar guns may impact the interpretation of maximum speed exposure data. Further, the Openfield software

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quantifies maximum velocity instantaneously (highest 10hz [100 ms] doppler velocity), whereas effort frequencies are determined according to the application of the minimum effort duration (dwell-time; 300 ms seconds herein). The effect of dwell time was not considered on near-to MS exposure counts and might be considered in future studies given its effect on exposure counts [21]. Finally, the case-study nature of this study limits generalizability, and the findings may be specific to the athletic culture and organisational settings.

**Conclusion**

Maximum speed exposure data are impacted by the determination approach, however the magnitude of the effect appears relatively trivial at the near-to maximum speed exposures considered more relevant for both performance gains and injury risk reduction purposes. Meaningful differences in lower speed exposures were observed, and are likely to be extrapolated for distance and time-above ≥80% and ≥85% maximum speed metrics. Given the current perceived importance of weekly near-to maximal speed exposures in applied practice [2,6] further work is necessary to confirm, refine or refute the available data, particularly in regards to injury risk associations. Moreover, further examination of speed exposure metrics and their measurement properties (i.e. sensitivity, validity) is necessary to inform and advance industry practice.

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**Declaration of interest statement**

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**References**

1. Faude, O., Rommers, N. & Rossler, R. Exercise-based injury prevention in football. *German Journal of Exercise and Sport Research* 48, 157–168 (2018).

2. McCall, A. *et al.* Exercise-Based Strategies to Prevent Muscle Injury in Male Elite Footballers: An Expert-Led Delphi Survey of 21 Practitioners Belonging to 18 Teams from the Big-5 European Leagues. *Sports Med* 50, 1667–1681 (2020).

3. Colby, M. J. *et al.* Repeated Exposure to Established High Risk Workload Scenarios Improves Non-Contact Injury Prediction in Elite Australian Footballers. *International journal of sports physiology and performance* 1–22 (2018) doi:10.1123/ijspp.2017-0696.

4. Hickey, J., Shield, A. J., Williams, M. D. & Opar, D. A. The financial cost of hamstring strain injuries in the Australian Football League. *British Journal of Sports Medicine* 48, 729–730 (2014).

5. Hägglund, M. *et al.* Injuries affect team performance negatively in professional football: an 11-year follow-up of the UEFA Champions League injury study. *British Journal of Sports Medicine* 47, 738–742 (2013).

6. Edouard, P. *et al.* Sprinting: a key piece of the hamstring injury risk management puzzle. *Brit J Sport Med* bjsports-2022-105532 (2022) doi:10.1136/bjsports-2022-105532.

7. Malone, S., Roe, M., Doran, D. A., Gabbett, T. J. & Collins, K. High chronic training loads and exposure to bouts of maximal velocity running reduce injury risk in elite Gaelic football. *J Sci Med Sport* 20, 250–254 (2017).

8. O'Connor, F. *et al.* Greater Association of Relative Thresholds Than Absolute Thresholds With Noncontact Lower-Body Injury in Professional Australian Rules Footballers: Implications For Sprint Monitoring. *International journal of sports physiology and performance* 1–25 (2019) doi:10.1123/ijspp.2019-0015.

9. Roe, G. *et al.* Validity of 10 HZ GPS and Timing Gates for Assessing Maximum Velocity in Professional Rugby Union Players. *International journal of sports physiology and performance* 1–14 (2016) doi:10.1123/ijspp.2016-0256.

10. Varley, M. C., Fairweather, I. H. & Aughey, R. J. Validity and reliability of GPS for measuring instantaneous velocity during acceleration, deceleration, and constant motion. *Journal of Sports Sciences* 30, 121–127 (2012).

11. Massard, T., Eggers, T. & Lovell, R. Peak speed determination in football: is sprint testing necessary? *Science and Medicine in Football* 00, 1–4 (2017).

12. Dillon, P., Norris, D., Siegler, J., Joyce, D. & Lovell, R. Determination of locomotor qualities in elite Australian Football: A pragmatic approach. *J Sports Sci* 39, 1445–1451 (2021).

13. Kyprianou, E. *et al.* A novel approach to assessing validity in sports performance research: integrating expert practitioner opinion into the statistical analysis. *Science and Medicine in Football* 00, 1–6 (2019).

14. Buchheit, M., Simpson, B. M., Hader, K. & Lacome, M. Occurrences of near-to-maximal speed-running bouts in elite soccer: insights for training prescription and injury mitigation. *Science and Medicine in Football* 5, 105–110 (2021).

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15. Brooks, M., E. *et al.* glmmTMB Balances Speed and Flexibility Among Packages for Zero-inflated Generalized Linear Mixed Modeling. *R J* 9, 378 (2017).

16. Hartig, F. *DHARMa: Residual Diagnostics for Hierarchical (Multi-Level / Mixed) Regression Models*. (2022).

17. Lenth, R. V. *emmeans: Estimated Marginal Means, aka Least-Squares Means*. (2021).

18. Buchheit, M., Settembre, M., Hader, K. & McHugh, D. Exposures to near-to-maximal speed running bouts during different turnarounds in elite football. (2022) doi:10.51224/srxiv.170.

19. Pyne, D. B., Gardner, A. S., Sheehan, K. & Hopkins, W. G. Positional differences in fitness and anthropometric characteristics in Australian football. *J Sci Med Sport* 9, 143–150 (2006).

20. Johnston, R. D., Black, G. M., Harrison, P. W., Murray, N. B. & Austin, D. J. Applied Sport Science of Australian Football: A Systematic Review. *Sports Med* 1–22 (2018) doi:10.1007/s40279-018-0919-z.

21. Varley, M. C., Jaspers, A., Helsen, W. F. & Malone, J. J. Methodological Considerations When Quantifying High-Intensity Efforts in Team Sport Using Global Positioning System Technology. *International journal of sports physiology and performance* 12, 1059–1068 (2017).

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**Figures**

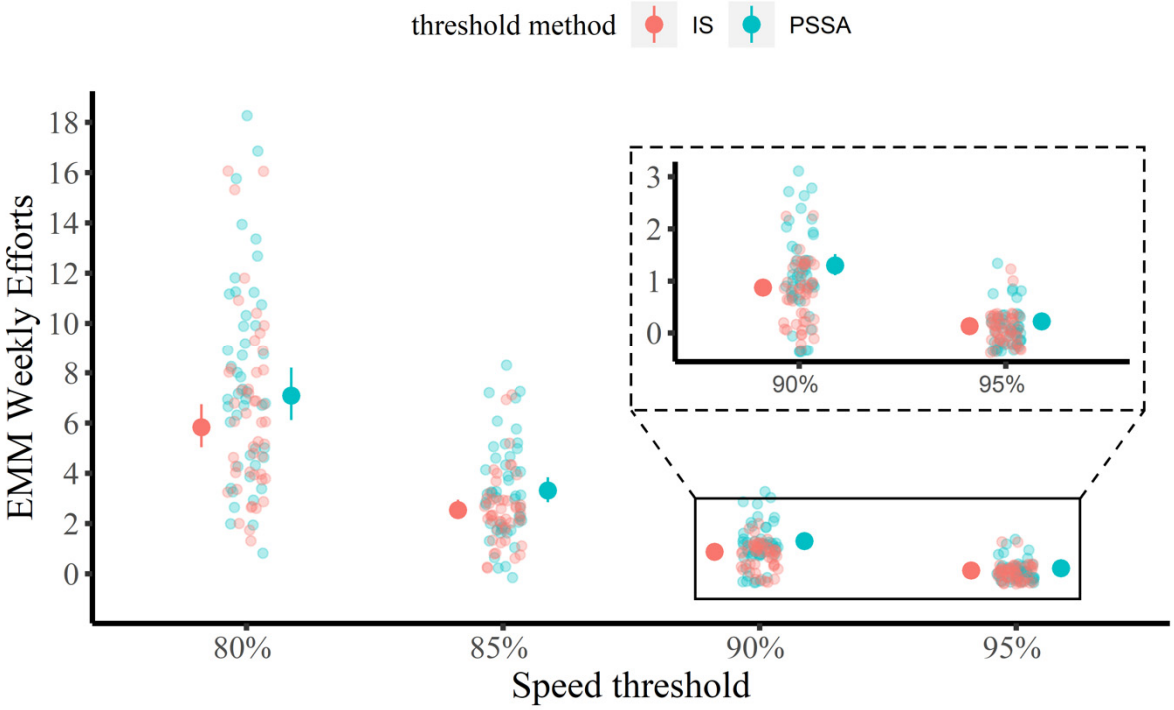


Figure 1

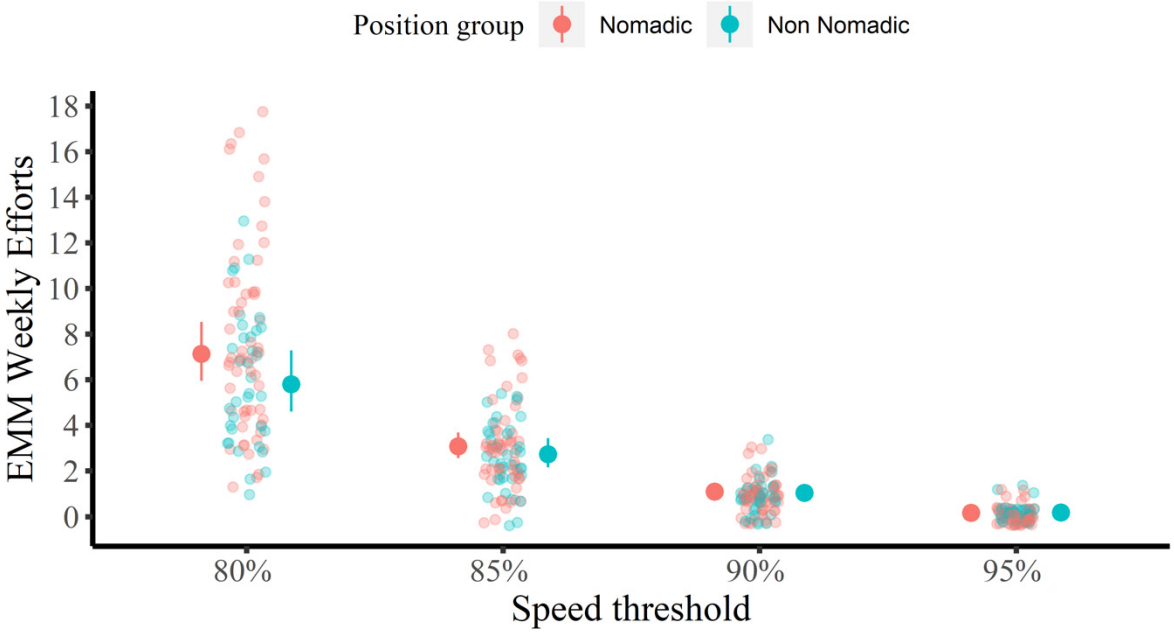


Figure 2

## Figure Captions

Figure 1: Weekly number of near-to maximum speed (MS) exposures according to determination method and threshold criteria. Error bars represent estimated marginal means (EMM) and corresponding 95% confidence intervals for average weekly exposures within the threshold category. The corresponding dots represent the individual average weekly estimates for each threshold method. IS = in-season monitoring; PSSA = pre-season sprint assessments.

Figure 2: Weekly number of near-to maximum speed (MS) exposures according to positional role and threshold criteria. Error bars represent estimated marginal means (EMM) and corresponding 95% confidence intervals for average weekly exposures within the threshold category. The corresponding dots represent the individual average weekly estimates for each threshold method.

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